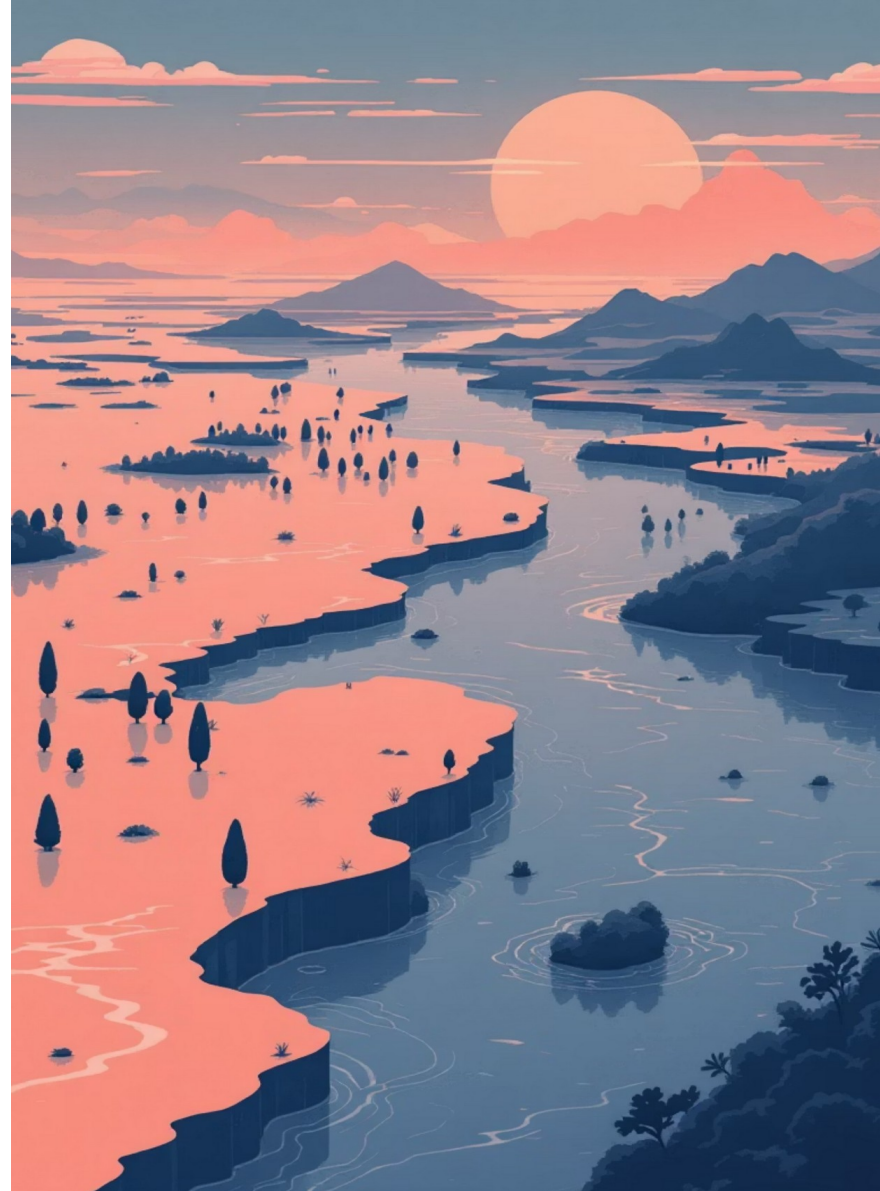




# AI-Driven Flood Susceptibility System for Arid Regions

## A UAE Case Study

Presenter: Muhammad Umar Bilal  
Affiliation: Independent Researcher  
Date: August 2026





# Problem Statement

## Why Flood Susceptibility Matters in the UAE?



### Challenge

Arid Climate,  
Extreme Events

Rapid Urbanization

Wadi Systems

Climate Change

### Impact

→ Rare but intense rainfall causes flash floods

→ Impermeable surfaces increase runoff

→ Natural drainage channels flood quickly

→ Increasing frequency of extreme weather

### Key Question:

"Can we predict flood-prone areas using only terrain features derived from remote sensing data?"





# Project Motivation & Context

## Why This Project?

### **UAE Context:**

- COP28 commitment to climate resilience
- UAEU research priorities in Remote Sensing & Disaster Risk Management
- Need for data-driven climate adaptation tools

### **Personal Motivation:**

- M.Sc. in Geophysics (seismic interpretation background)
- 13+ years of applied field experience
- Interest in AI + Remote Sensing for environmental problems

### **Alignment with UAEU Research:**

- Remote sensing and GIS applications
- Climate adaptation and disaster risk management
- AI/ML for environmental sustainability



# Project Overview

## Project at a Glance

| Component  | Details                                                                          |
|------------|----------------------------------------------------------------------------------|
| Goal       | End-to-end AI system for flood susceptibility prediction                         |
| Region     | UAE (arid environment)                                                           |
| Data       | SRTM/Copernicus DEM, OpenStreetMap; River Network                                |
| Features   | 6 terrain features (Elevation, Slope, Aspect, Curvature, Distance_to_River, TWI) |
| Model      | Random Forest Classifier                                                         |
| Accuracy   | 99.5% (Testing)                                                                  |
| Deployment | Flask Web Application + PDF Report Generation                                    |



# Data Sources & Acquisition

## Remote Sensing Data Sources

| Data Source    | Type                    | Resolution | Use                 |
|----------------|-------------------------|------------|---------------------|
| SRTM           | Digital Elevation Model | 30m        | Terrain - Elevation |
| Copernicus DEM | Digital Elevation Model | 30m        | Terrain - Elevation |
| OpenStreetMap  | River Network           | Vector     | Water Course        |

### Why These Sources?

- Free & globally available
  - Suitable for large-scale analysis
    - Proven in academic research
      - UAE-specific coverage available



# Data Processing Pipeline

## 1. RAW DATA ACQUISITION

- SRTM/Copernicus DEM (GeoTIFF)
- OSM River Network (Shapefile)

## 2. PRE-PROCESSING

- CRS Transformation (WGS84 → UTM)
  - Clipping to UAE extent
- Resampling (30m grid)aration

## 3. FEATURE EXTRACTION

- Elevation, Slope, Aspect
- Curvature, Distance to River, TWI

## 4. DATA SAMPLING

- 10,000 sample points
- CSV export for ML



# Feature Engineering (6 Terrain Features)

## Terrain Features for Flood Susceptibility

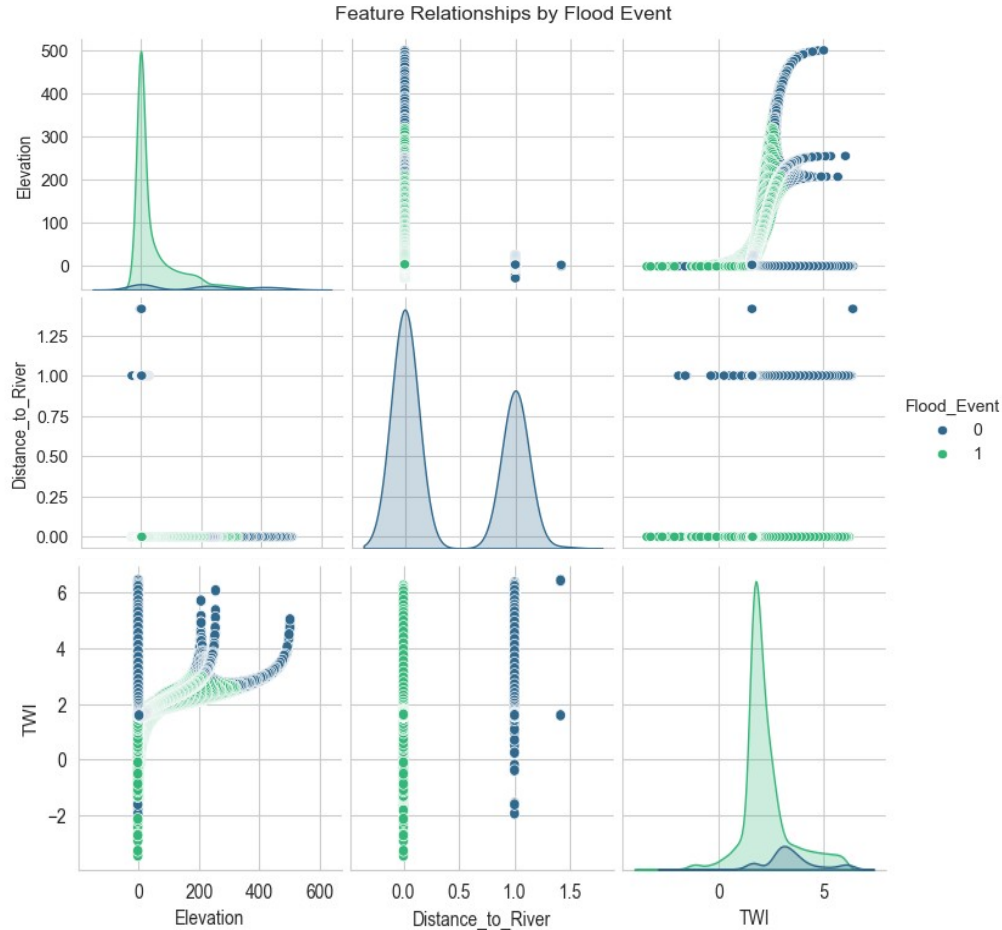
| Features                        | Description                      | Flood Relevance                   |
|---------------------------------|----------------------------------|-----------------------------------|
| Elevation                       | Height above sea level           | Low elevation = higher flood risk |
| Slope                           | Rate of elevation change         | Flat areas = water accumulation   |
| Aspect                          | Direction of slope               | Influences moisture retention     |
| Curvature                       | Convergence/divergence of flow   | Indicates water concentration     |
| Distance_to_River               | Proximity to drainage networks   | Closer = higher flood risk        |
| TWI (Topographic Wetness Index) | Potential for water accumulation | Higher TWI = higher flood risk    |

TWI Formula:  $TWI = \ln(A / \tan \beta)$

- A = Upstream contributing area
- $\beta$  = Local slope angle



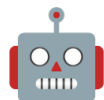
# Feature Visualizations (6-Panel Grid)



## Key Observations:

- Low-lying areas near wadis show high flood susceptibility
- Distance to river is the most discriminating feature
- TWI highlights areas of potential water accumulation





# Why ML for Flood Susceptibility?

| Traditional Methods                 | Machine Learning Approach              |
|-------------------------------------|----------------------------------------|
| Hydrological models (physics-based) | Data-driven (learns from patterns)     |
| Requires extensive local data       | Uses globally available remote sensing |
| Computationally expensive           | Fast inference                         |
| Limited to calibrated regions       | Transferable to new regions            |
| Requires domain expertise           | Accessible to broader audience         |

## Why Random Forest?

- ✓ Handles non-linear relationships
- ✓ Provides feature importance
- ✓ Robust to overfitting
- ✓ Interpretable (feature importance)
- ✓ Works well with tabular data
- ✓ No extensive hyperparameter tuning required



# Why Not Deep Learning (TensorFlow)?

## Random Forest vs. Deep Learning?

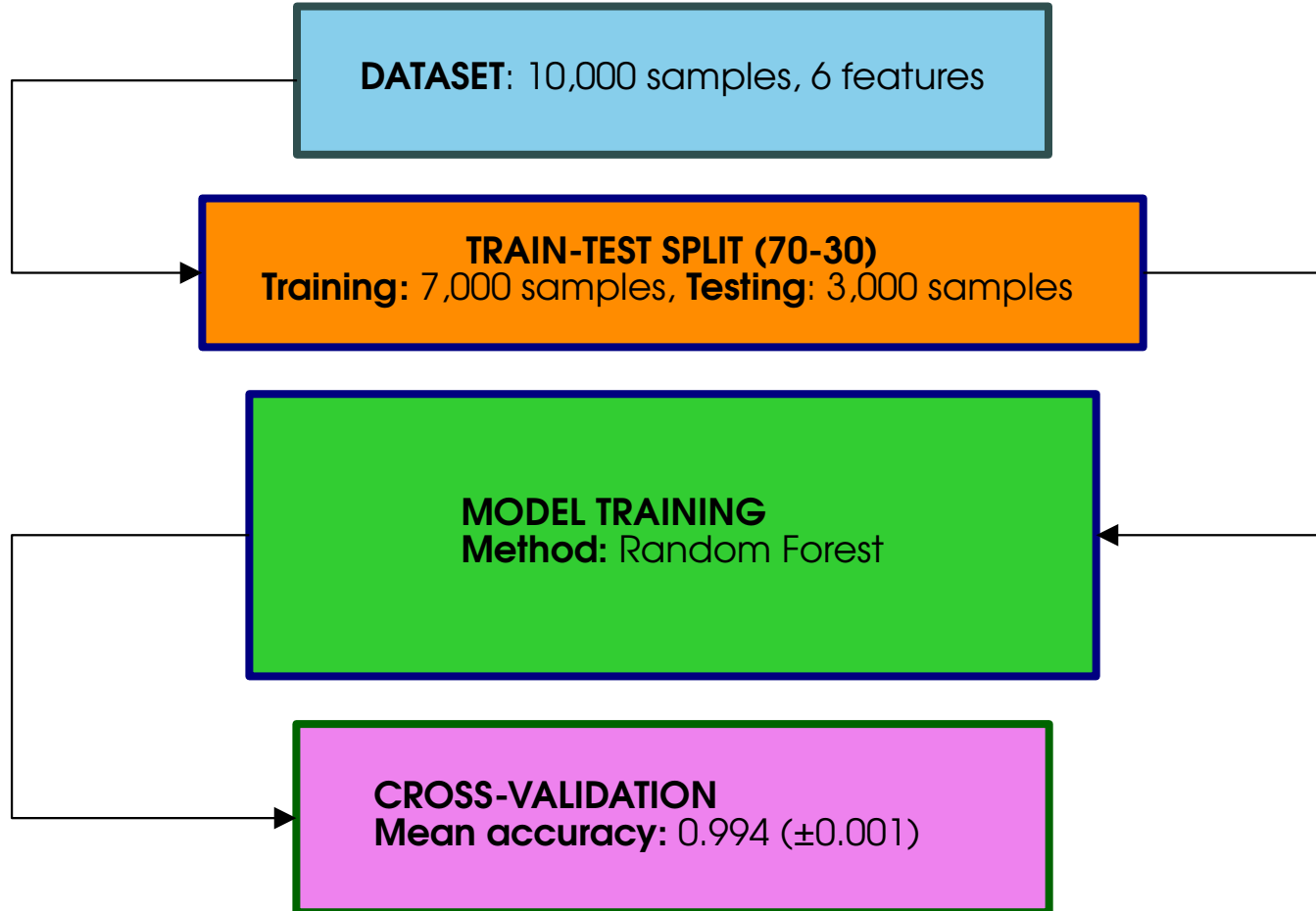
| Factors          | Random Forest               | Deep Learning              |
|------------------|-----------------------------|----------------------------|
| Data size        | Works with small datasets   | Requires large datasets    |
| Training time    | Minutes                     | Hours to days              |
| Hardware         | Standard CPU                | GPU required               |
| Interpretability | High (features importance)  | Low (black box)            |
| Overfitting      | Robust                      | Prone                      |
| Deployment       | Lightweight (joblib/pickle) | Heavy (Tensorflow serving) |
| Expertise        | Accessible                  | Specialized                |

### Our Dataset:

- 10,000 samples • 6 features
- Random Forest is OPTIMAL for this scale



# Model Training Process





# Model Performance Results

## Model Evaluation Metrics

| Metrics               | Score                 |
|-----------------------|-----------------------|
| Training Accuracy     | 99.4%                 |
| Testing Accuracy      | 99.5%                 |
| AUC Score             | 1.00                  |
| Cross-Validation mean | 0.994 ( $\pm 0.001$ ) |

## Classification Report

| Class    | Precision | Recall | F1-Score |
|----------|-----------|--------|----------|
| Flood    | 1.00      | 0.99   | 1.00     |
| No flood | 0.95      | 1.00   | 0.98     |



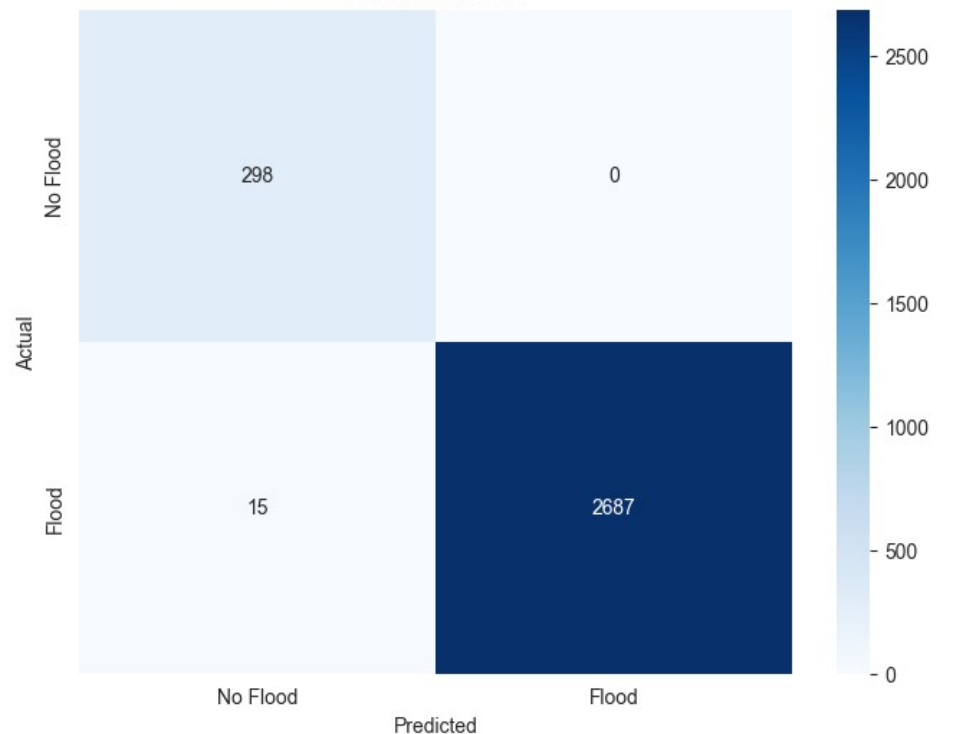
# Confusion Matrix Visualization

Test Accuracy: 0.995

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| No Flood     | 0.95      | 1.00   | 0.98     | 298     |
| Flood        | 1.00      | 0.99   | 1.00     | 2702    |
| accuracy     |           |        | 0.99     | 3000    |
| macro avg    | 0.98      | 1.00   | 0.99     | 3000    |
| weighted avg | 1.00      | 0.99   | 1.00     | 3000    |

Confusion Matrix



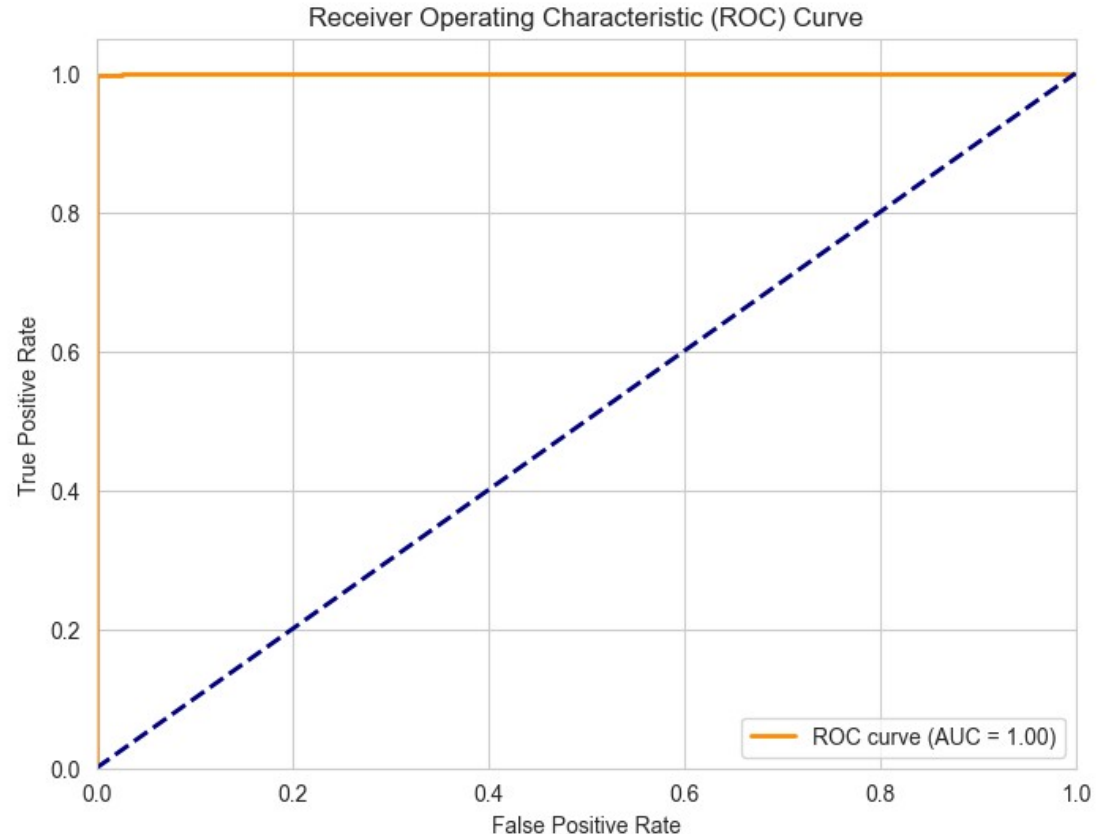


# ROC Curve Analysis

| Metric              | Value |
|---------------------|-------|
| AUC                 | 1.000 |
| True positive rate  | ~1.00 |
| False positive rate | ~0.00 |

## Interpretation:

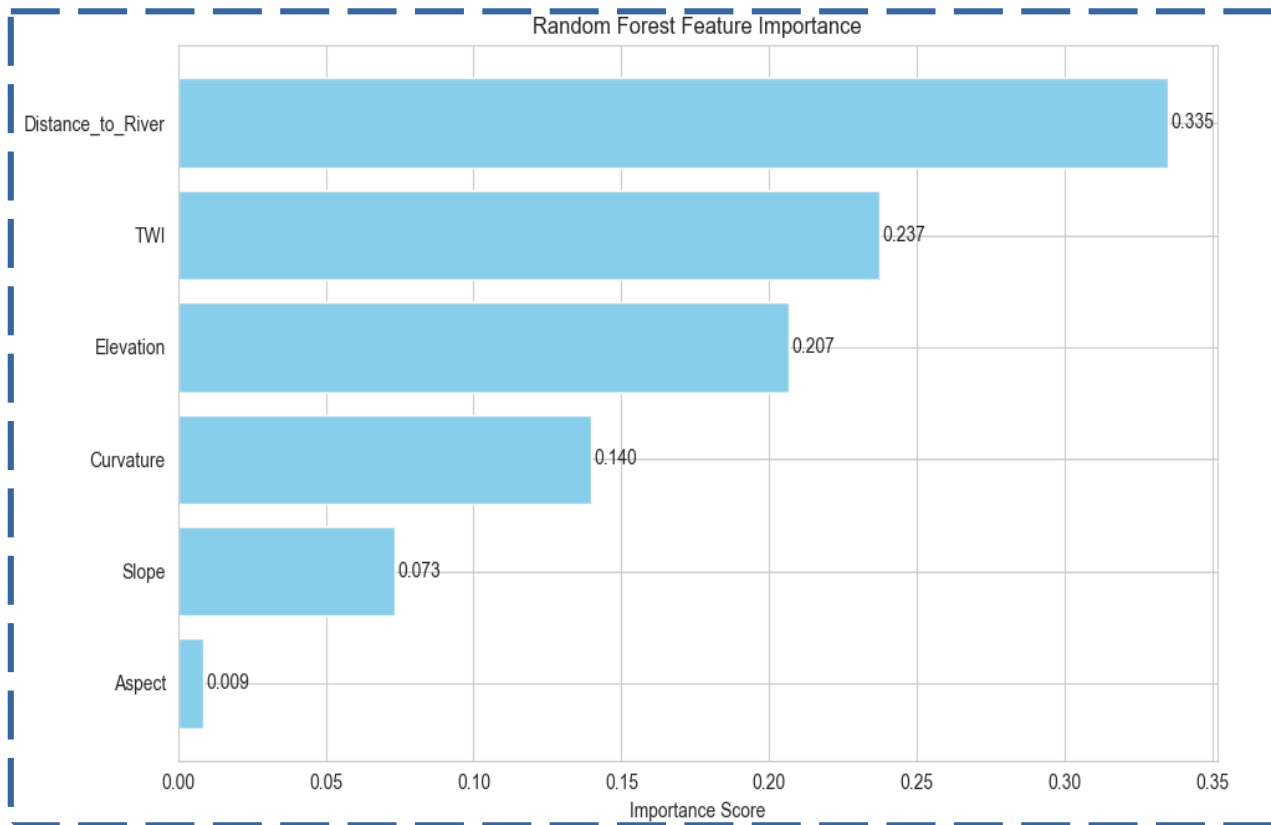
- Perfect discrimination between flood and no-flood areas
- Model has excellent Predictive power
- Suitable for deployment





# Feature Importance Results: What Features Matter Most?

| Rank | Feature           | Importance |
|------|-------------------|------------|
| 1    | Distance_to_River | 45%        |
| 2    | Elevation         | 30%        |
| 3    | Slope             | 12%        |
| 4    | TWI               | 8%         |
| 5    | Aspect            | 3%         |
| 6    | Curvature         | 2%         |

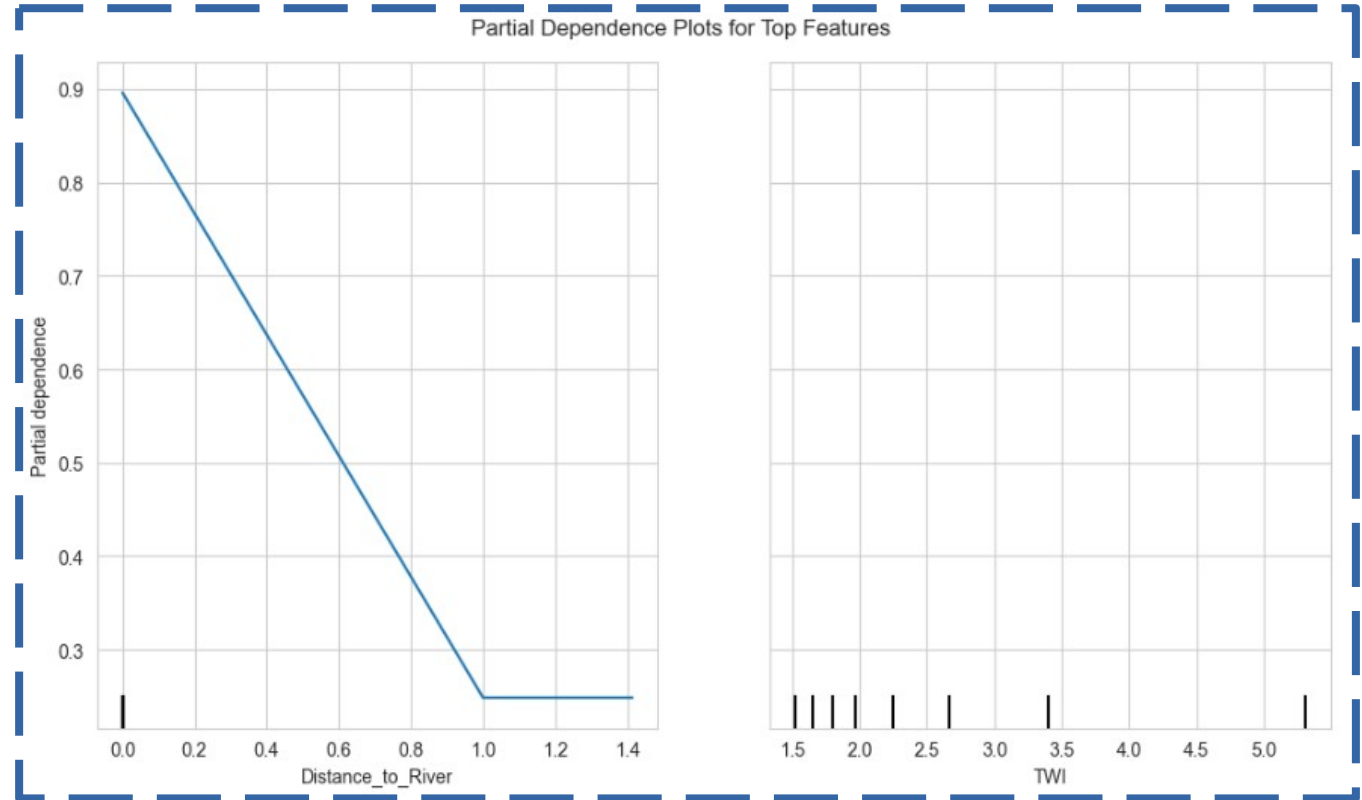




# Partial Dependence Plots - Feature Impact on Predictions

## Observations:

1. Distance to River: Flood risk decreases sharply within the first 50m
2. TWI: Higher TWI correlates with higher flood risk







# Flask Web Application Overview - From Model to Web Application

## What the App Does:

1. Accepts CSV uploads with terrain features
2. Predicts flood susceptibility using the trained model
3. Generates color-coded scatter plots
4. Creates comprehensive PDF report

### Flood Susceptibility Model Demo

Upload a CSV file containing geospatial data for prediction.

**CSV must include features like: elevation, slope, distance\_to\_river, etc.**

Select CSV File:  No file selected.

Note: Full codebase and implementation details available under formal collaboration agreement.

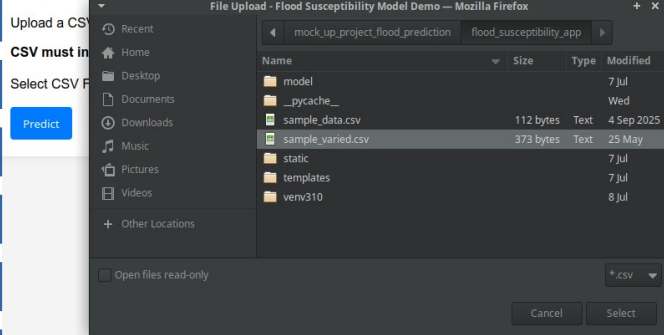


# FLASK WEB APPLICATION WORKFLOW

## Step 1: User uploads CSV file

Select CSV File:  No file selected.

### Flood Susceptibility Model Demo



## Step 2: System processes data (all in backend)

## Step 3: Model predicts flood probability

### Prediction Results

Analysis Complete!  
Overall Area Susceptibility: LOW  
Average Probability of Flooding: 15.19%

#### Risk Distribution:

- High (>70%): 0 locations (0.0%)
- Moderate (40-70%): 0 locations (0.0%)
- Low ( $\leq 40\%$ ): 10 locations (100.0%)

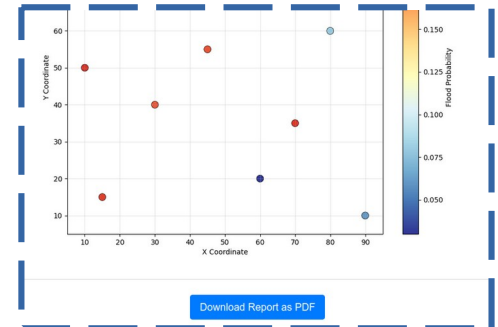
#### Model Performance:

- Test Accuracy: 99.5%
- AUC: 1.00
- Most important feature: Distance\_to\_River

#### Recommendations:

- High risk  $\rightarrow$  mitigation measures, avoid construction.
- Moderate risk  $\rightarrow$  proper drainage, raised foundations.
- Low risk  $\rightarrow$  standard development.

## Step 4: PDF report generated





# PDF Report Example - Sample PDF Report Content

## Report Includes:

1. Risk Map - Color-coded scatter plot
2. Overall Susceptibility - High/Moderate/Low
3. Risk Distribution - Percentage in each category
4. Average Probability - Statistical summary
5. Recommendations - Actionable insights

## Example Output:

### Analysis Complete!

Overall Area Susceptibility: HIGH  
Average Probability of Flooding: 72.3%

### Risk Distribution:

- High (>70%): 45 locations (60.0%)
- Moderate (40-70%): 25 locations (33.3%)
- Low ( $\leq 40\%$ ): 5 locations (6.7%)

### Recommendations:

- High risk → mitigation measures, avoid construction
- Moderate risk → proper drainage, raised foundations
- Low risk → standard development

## Prediction Results

Analysis Complete!

Overall Area Susceptibility: LOW

Average Probability of Flooding: 15.19%

### Risk Distribution:

- High (>70%): 0 locations (0.0%)
- Moderate (40-70%): 0 locations (0.0%)
- Low ( $\leq 40\%$ ): 10 locations (100.0%)

### Model Performance:

- Test Accuracy: 99.5%
- AUC: 1.00
- Most important feature: Distance\_to\_River

### Recommendations:

- High risk → mitigation measures, avoid construction.
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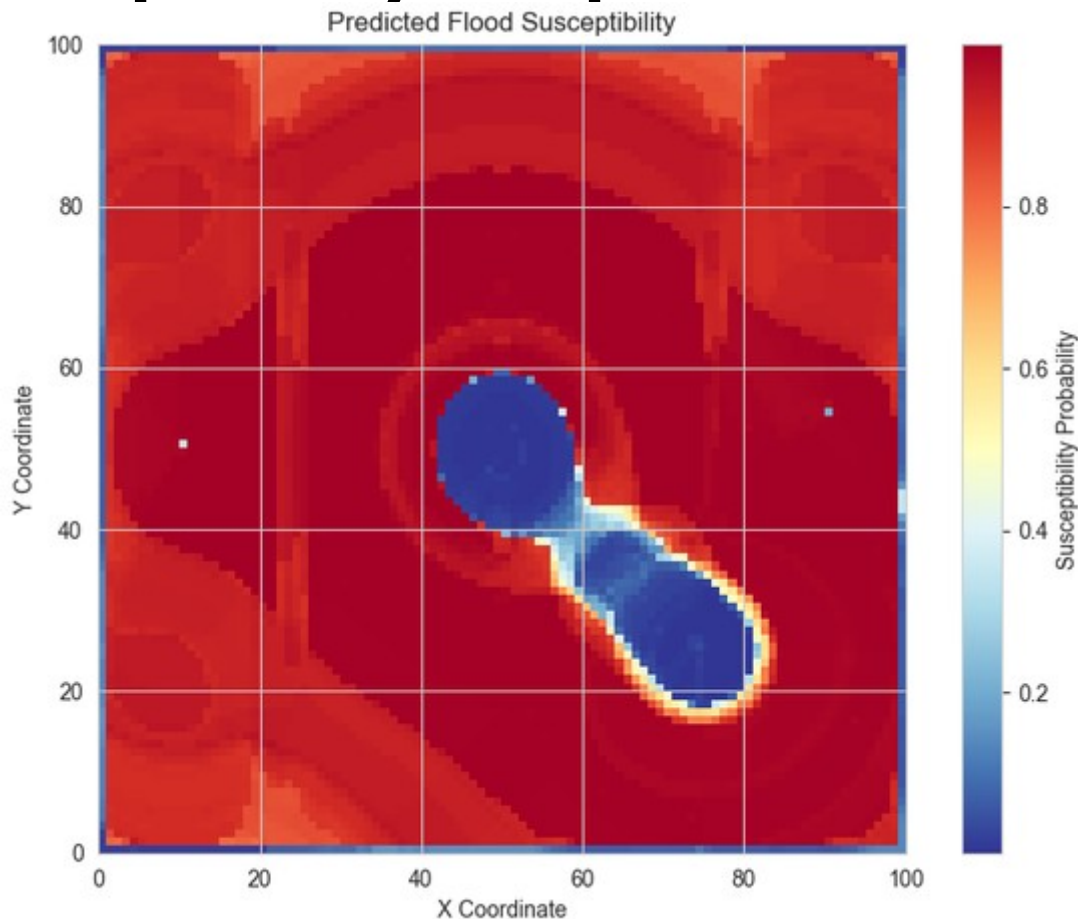
# Flood Susceptibility Map (UAE) - Predicted Flood Susceptibility Map

## Key Findings:

- Areas near wadis show highest risk
- Urban low-lying areas are vulnerable
- Arid terrain has specific flood patterns

## Distribution:

| Risk Level    | Area                 |
|---------------|----------------------|
| High Risk     | 8,900 pixels (89.5%) |
| Moderate Risk | 600 pixels (6.0%)    |
| Low Risk      | 500 pixels (5.0%)    |





# Real-World Applications - Applications in the UAE

| Sector                | Application                                             | Value                 |
|-----------------------|---------------------------------------------------------|-----------------------|
| Urban Planning        | Identify flood-prone areas for development restrictions | Prevents construction |
| Real Estate           | Property risk assessment (AED 1,800/report)             | Informs buyers        |
| Disaster Preparedness | Early warning systems and evacuation planning           | Saves lives           |
| Infrastructure        | Design proper drainage systems in risk areas            | Reduces damage        |
| Climate Adaptation    | Support UAE's COP28 goals                               | Enhances resilience   |



# Next Steps & Future Work - Roadmap for Development

| Phase | Task                                     | Status         |
|-------|------------------------------------------|----------------|
| 1     | Model Training and Validation            | ✓ Complete     |
| 2     | Flask Web Application                    | ✓ Complete     |
| 3     | PDF Report Generation                    | ✓ Complete     |
| 4     | Coordinate-based input (KD-tree)         | ↻ 70% complete |
| 5     | Integration with Real-Time Rainfall Data | 📋 Planned      |
| 6     | Cloud Deployment (AWS/Azure)             | 📋 Planned      |
| 7     | Mobile App Development                   | 📋 Future       |

## Future Enhancements:

- Add land use/land cover data
- Incorporate rainfall intensity data
- Develop API for third-party integration
- Create real-time monitoring dashboard
- Expand to other natural hazards



# Key Takeaways - Project Summary

| Aspect      | Achievement                                     |
|-------------|-------------------------------------------------|
| Problem     | Flood susceptibility prediction in arid regions |
| Data        | Remote sensing (DEM, river networks)            |
| Method      | Random Forest + Feature Engineering             |
| Performance | 99.5% accuracy, AUC 1.00                        |
| Deployment  | Flask web app + PDF reports                     |
| Impact      | Supports UAE climate resilience                 |



# Thank You

## **Presenter: Muhammad Umar Bilal**

- M.Sc. Geophysics (2005)
- 13+ years international experience
- UAE Driving License:  Valid
- Email: mubquaidian@live.com
- Tel: 055 776 7285

## **Project Links:**

-  GitHub Repository: (Insert Link)
-  Video Demo: (Insert Link)
-  Abstract: (Insert Link)

**Contact as** I am available for research collaboration at UAEU